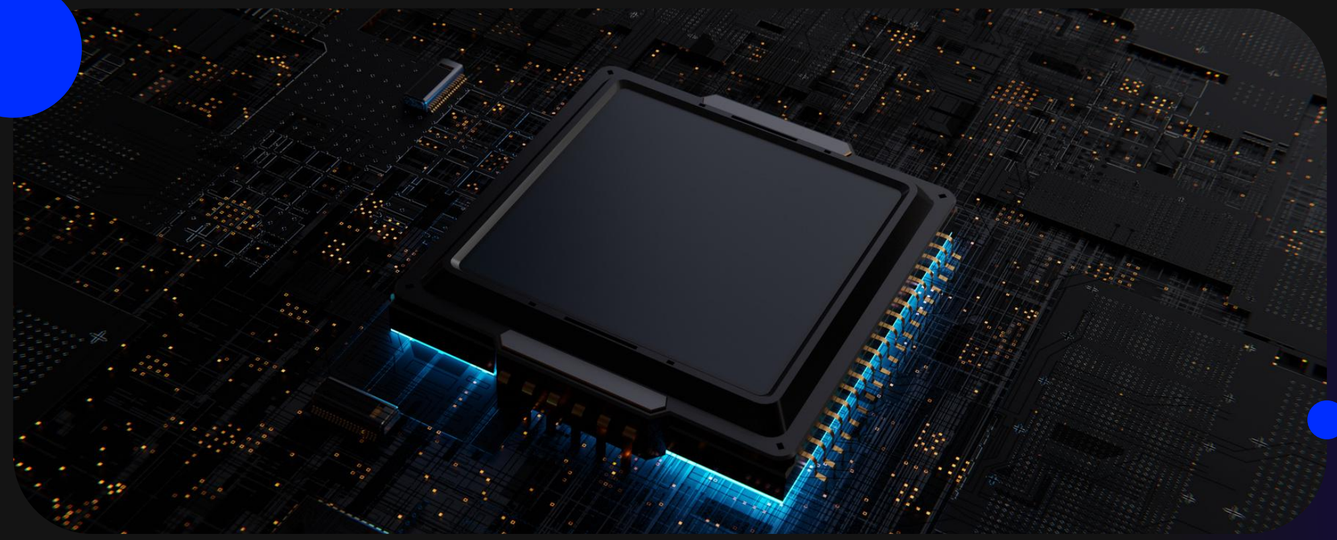
Quantum Computing: transforming the future of computing



By: Anniha Dorce

Introduction to Quantum Computing

What is a Quantum Computer?

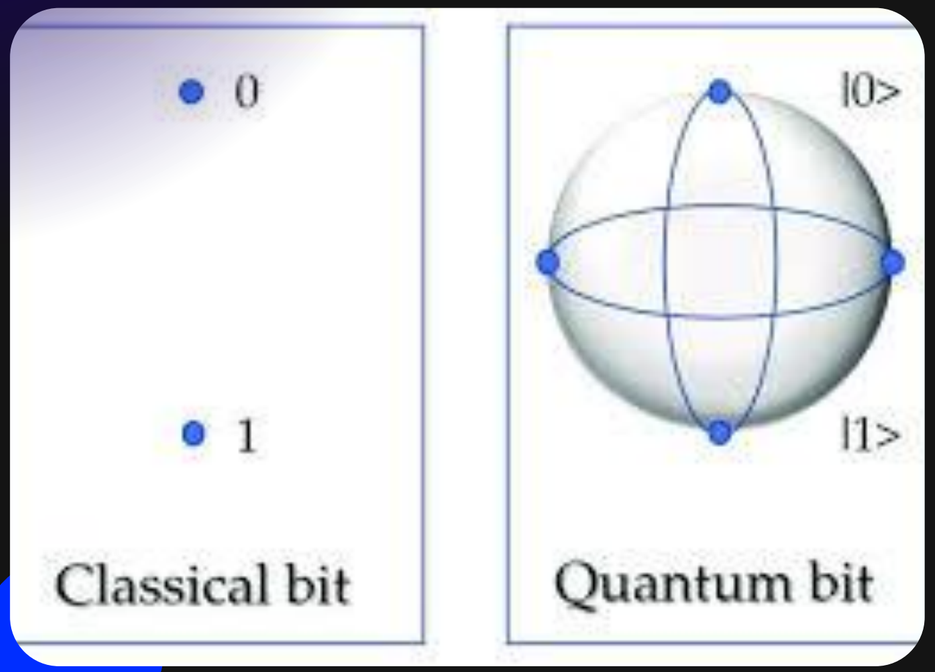
A quantum computer is an advanced computer that uses the principles of quantum mechanics to solve certain problems far beyond the speed and capability of classical computers. Quantum computers can solve problems in minutes or hours that would take classical computers centuries.



The difference between Quantum and Classic

Quantum computers can solve problems in minutes or hours that would take classical computers centuries. The Key difference is how they process information. Classical computers use bits. while Quantum computers use qubits.

Qubits & Key Principles



A qubit (quantum bit) is the basic unit of quantum information. Qubits can be in state Ket 0, Ket 1, or both at once . This is a property called superposition. Entanglement links qubits so that changing one instantly affects the other, no matter the distance. Quantum gates stand as the fundamental building blocks that enable the transformation of quantum information. Quantum gates manipulate qubits by rotating their states, similar to how classical logic gates flip bits.

Why do these principles matter: They allow quantum computers to process information in fundamentally different and new ways classical computers can't.

Superposition

Quantum Entanglement

Quantum Gates

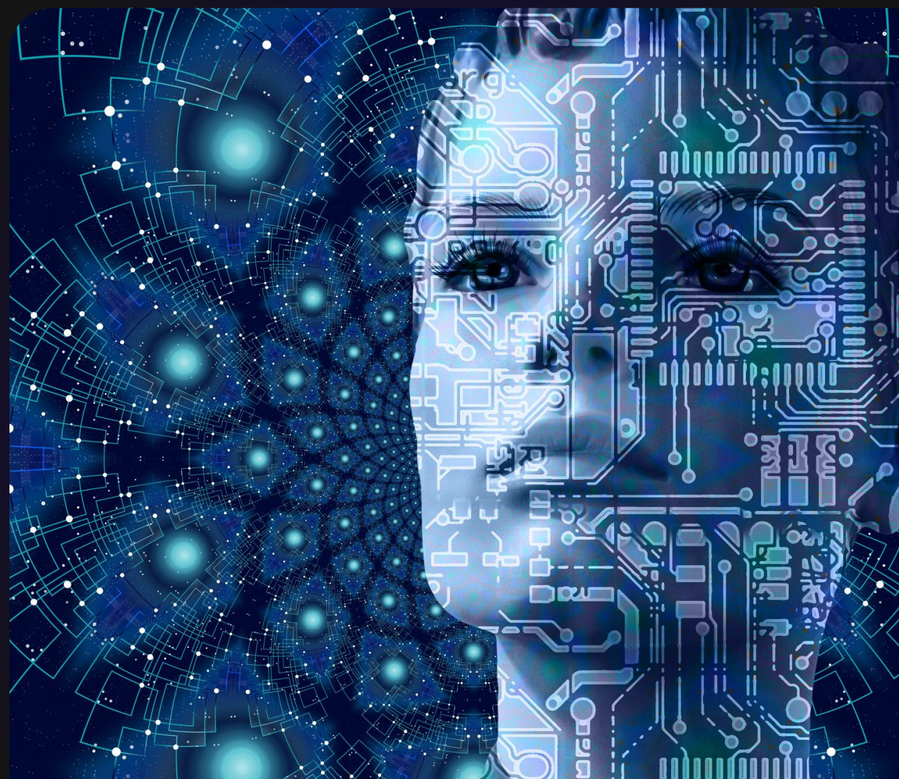
Intended & Unintended Consequences

Intended Consequences

- Faster problem-solving for complex tasks
- New scientific discoveries
- Stronger AI and machine-learning models

Unintended Consequences

- Quantum computers could break RSA and ECC encryption, threatening national security, banking, and personal data.
- Unequal access to these computers raises ethical concerns.
- Potential misuse regarding surveillance or warfare



Fundamental Limitations

We are still in the early Days of Quantum Computing

Most systems require extreme cryogenic cooling and can only maintain quantum states briefly.

Quantum computers excel only at specific types of problems; they are not universally faster than classical computers.

The quantum computers we have today are mostly used for research

Prohibitively Expensive

Quantum hardware, talent, and supply chains are extremely expensive, slowing widespread adoption.

Inadequate workforce

There aren't enough skilled workers in quantum engineering, physics, and software to meet demand.

Hardware Development

Creating reliable qubits and control hardware is complex

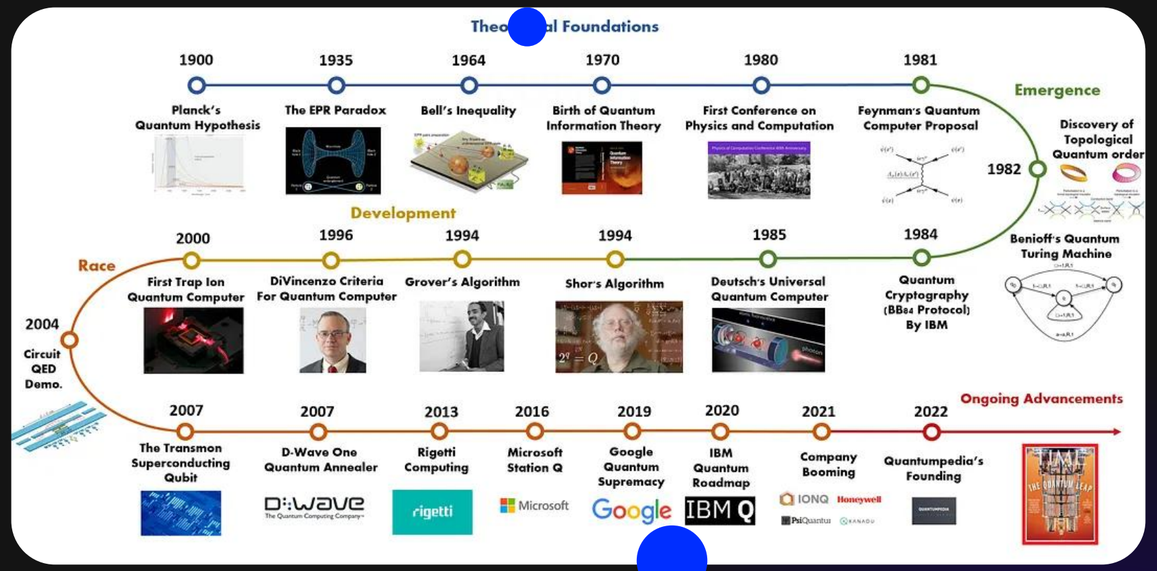
Software Development

Quantum algorithms and programming tools are still early-stage and limited

Transformational Impact & Future Outlook

Quantum computers could revolutionize the future of technology; they use quantum mechanics to solve complex multidimensional problems at a fraction of the time it would take classic computers. Classical computing is reaching physical limits (Moore's Law), but quantum computing opens new possibilities.





Pharmaceuticals: Quantum computers could simulate drug molecules digitally, reducing years of lab testing.

Healthcare: Quantum machine learning may detect disease patterns hidden in high-dimensional medical data.

Chemistry: They can model molecular interactions, such as photosynthesis, to improve energy efficiency research.

Machine Learning

Real-World Applications

Machine Learning: Quantum algorithms may speed up training and optimization.

Finance: Useful for risk analysis, portfolio optimization, and market modeling.

Cybersecurity: Quantum computers threaten current encryption but also enable quantum-safe cryptography.

Thank You

Citations Page

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